Sahaj Bhadja

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Summary

Software engineer with 2+ years building Python and Java services, REST APIs, cloud infrastructure, and reliable backend systems. Experienced in C++, AWS, scientific computing, testing, debugging, CI/CD, performance optimization, documentation, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs and Azure SQL services for 500+ users, profiling indexed queries and backend access paths to reduce production latency from 1.2 seconds to 350 milliseconds.
- Orchestrated 5,000+ monthly jobs through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving system observability, maintainability, operational support, failure recovery, and systematic production troubleshooting responsiveness.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services handling 10,000+ daily requests, integrating secure document ingestion, authentication, structured status updates, and automated workflows while systematically debugging production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress tracking, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable Python and PyTorch pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, structured evaluation, performance profiling, validation checks, and systematic model debugging.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, SQL integrations, and ETL workflows using AWS Glue, S3, and REST APIs, improving response times 30% and data-processing efficiency 68% across production cloud applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating defect diagnosis, release verification, and reliable cross-platform software deployments across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular Node.js services across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, coordinating 2 million+ Azure SQL records through clearly documented secure interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, deployment consistency, and operational visibility.

Skills

Languages & Computing: Python, C++, Java, JavaScript, TypeScript, SQL, PyTorch, scientific computing, object-oriented programming, data structures, algorithms
Backend & Systems: Flask, FastAPI, Node.js, REST APIs, microservices, WebSockets, multithreading, backend services, distributed workflows, performance optimization
Cloud & Data: AWS, Azure, AWS Glue, S3, AWS RDS, Azure SQL, PostgreSQL, MySQL, MongoDB, Docker, Kubernetes, cloud services, data processing
Quality & Delivery: Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, functional testing, automated testing, debugging, profiling, logging, monitoring, code review, documentation, build automation, production support, Agile

Projects

Big-Brain | Live | *LLaMA-2, RAG, FAISS, PostgreSQL, REST APIs, Clerk*
- Built an authenticated RAG application processing 50,000+ unstructured documents with LLaMA-2, FAISS, PostgreSQL, and REST APIs, combining semantic retrieval, conversational context, access controls, and real-time collaboration workflows.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular Java and TypeScript services.

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms, Performance Analysis*
- Built a C++ simulator implementing six scheduling algorithms with deterministic workloads, edge-case validation, Gantt visualization, and performance analysis, reducing average turnaround time 25% across 1,000+ process configurations.